

JATIN SATYAM

MS Robotics & Autonomous Systems @ ASU · Robotics · Mechatronics · Automation

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SUMMARY

Mechanical engineer turned robotics engineer - MS Robotics and Autonomous Systems (Systems Engineering) at Arizona State University - building ROS 2 pipelines spanning Bayesian occupancy mapping, terrain-relative navigation, SLAM drones, LQR optimal control, and LLM-driven manipulation on physical robot arms.

EXPERIENCE

Volunteer Research Assistant

Aug 2026 – Present

Robotics Systems Instructional Lab, Arizona State University

- Building a two-robot semiconductor wafer-handling cell (Dobot M1 Pro SCARA + myCobot Pro 600, Allen-Bradley Micro850 PLC, pneumatic vacuum grippers, photoelectric sensors) and developing its digital twin in Gazebo on ROS 2 Humble, then NVIDIA Isaac Sim

Mechatronics Engineer Intern

Jan 2025 – Jul 2025

Paras Defence & Space Technologies Ltd., Navi Mumbai, India · full-time at graduate-engineer-trainee scope

- Designed and iterated Gifford-McMahon cryocooler mechanisms in SolidWorks and NX, refining moving-assembly tolerances and interfaces for reliable cold-head displacer motion and manufacturability
- Led performance testing of 3 cryogenic cooling assemblies across 25+ instrumented runs, characterizing thermal losses and translating findings into mechanism-level design changes that cut prototype rework by 12%

R&D Engineer Intern

May 2023 – Aug 2023

TriQual Cryogenics Pvt Ltd, Navi Mumbai, India

- Supported mechanical design and vibration-control features for Stirling-cycle cryocoolers operating near 70 K, working alongside senior engineers on reciprocating assemblies

PROJECTS

Aerial-Ground Cooperative Autonomy on Mars Terrain · ROS 2 · Gazebo · Python ASU - SES 598 Space Robotics & AI · 2026 · NumPy

- Built a 2-robot autonomy pipeline on real HiRISE Jezero Crater terrain in ROS 2 and Gazebo: a UAV surveys and builds a Bayesian log-odds occupancy grid (0.25 m/cell) that a 6-wheel rover navigates with A* replanning and terrain-relative navigation
- Implemented a vectorized terrain-relative navigation matcher - likelihood-field scoring of K=294 pose candidates at 2 Hz over 36 lidar rays - up to 24% lower localization error than odometry in feature-dense terrain (sparse-coverage limits documented)

Autonomous Drone Search, Map & Precision Landing · ROS 2 · PX4 · RTAB-Map · ASU - SES 598 Space Robotics & AI · 2026 · OpenCV

- Programmed a 9-state autonomous drone mission (PX4 SITL + ROS 2): search for rock formations, identify ArUco markers while orbiting, select the tallest cylinder, and land on top - perfect score across 3/3 trials (150/150 including extra credit)
- Ran RTAB-Map RGB-D SLAM live during flight, producing a 43,283-vertex textured 3D mesh of the environment from 78 optimized poses

SKILLS

Robotics, Controls & Autonomy: ROS 2, Gazebo, PX4 SITL, Forward/Inverse Kinematics, Motion Planning, A* Path Planning, Bayesian Log-Odds Occupancy Mapping, Terrain-Relative Navigation, PID, LQR, State-Space Modeling

Perception & Computer Vision: RTAB-Map (RGB-D SLAM), OpenCV, ArUco Detection, Point Cloud Processing

Programming & Software Tools: C++, Python (NumPy, SciPy, OpenCV), C, MATLAB, Linux, Git & GitHub

CAD & Mechanical Design: SolidWorks, Siemens NX, GD&T, DFM, Tolerance Analysis

Embedded & Electronics: Sensor Integration (LiDAR, Depth Cameras, IMU, Encoders), Microcontrollers, I2C, SPI, UART, CAN

EDUCATION

M.S. Robotics and Autonomous Systems (Systems Engineering), Arizona State University · GPA 3.67/4.0 Expected May 2027

Coursework: Robotic Systems I & II - Space Robotics & AI - Mechatronics Device Innovation - Advanced System Modeling & Control - Multi-Robot Systems (in progress)

B.E. Mechanical Engineering, University of Mumbai

2020 – 2024

CERTIFICATIONS

MATLAB Onramp - MathWorks, 2026 · Simulink Onramp - MathWorks, 2026